Parallel-Agent Reinforcement Learning and Agent Swarms

Deep Dive into Kimi K3 — Chapter 26

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Foundation Books Project

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Goal: type a swarm episode as a dynamic fork–join graph, train only the orchestrator against frozen workers, scaffold delegation with an annealed reward, count work and critical steps separately, and join evidence without losing provenance.

Roadmap

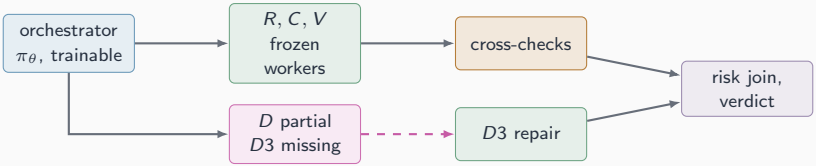
Typed graphs and a frozen-worker boundary

Reward and the step ledger

Evidence, experiments, and handoffs

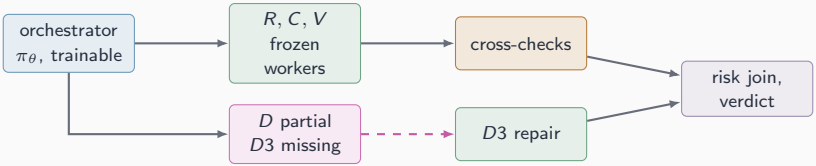
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Fork, repair, and join under one verifier



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- Only orchestrator action tokens enter the gradient; worker packets are observations.

[REPORTED: K2.5 Sec. 3]

Reward and the step ledger

Scaffolded reward, separate ledgers

- $r_{\mathrm{PARL}} = \lambda_1 r_{\mathrm{parallel}} + \lambda_2 r_{\mathrm{finish}} + r_{\mathrm{perf}}$, with λ_1, λ_2 annealed to zero. [REPORTED: K2.5 Sec. 3]

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A step proxy is not wall-clock time; K2.5's 3×–4.5× speedups are a source-reported setting.

Evidence, experiments, and handoffs

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The studio ends in a supported *not ready* verdict. Next: Chapter 27, Structured Generation, XTML, and Walle.